

QUANTUM-RESISTANT CYBER SECURITY APPARATUS

PROJECT REPORT

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CHAPTER 1 – INTRODUCTION

1.1 Project Description

The Quantum-Resistant Cyber Security Apparatus is a proposed secure communication system that combines encryption with image steganography. The system is designed to protect sensitive information by encrypting the data and then concealing the encrypted payload inside a carrier image. The present project phase focuses on the problem definition, literature study, system requirements, system design, architecture and workflow.

1.2 Problem Statement

Conventional security solutions mainly protect the contents of information through encryption, but the presence of encrypted data can itself attract attention. Steganography can conceal the existence of communication, but by itself it does not provide strong protection for the hidden information. Therefore, a combined approach is proposed in which encrypted information is hidden inside a lossless image.

1.3 Objectives

- Develop a secure communication concept combining cryptographic protection and steganography.
- Encrypt sensitive text or image information before embedding.
- Hide the encrypted payload using LSB image steganography.
- Provide integrity and validation mechanisms in the proposed workflow.
- Design a modular architecture that can later support post-quantum cryptographic algorithms.
- Prepare the system for future implementation and testing.

1.4 Scope

The current phase covers system analysis, requirements, architecture, data flow and design. Full software implementation, hardware integration and experimental validation are planned for subsequent phases.

CHAPTER 2 – LITERATURE SURVEY

2.1 Encryption and Secure Communication

Encryption converts readable information into an encoded form so that unauthorized users cannot understand the original data. Cryptographic protection is an essential layer for confidentiality and secure communication.

2.2 Image Steganography

Image steganography hides information inside a digital image. In LSB steganography, selected least significant bits of image pixels are modified to store the payload while attempting to preserve the visual appearance of the carrier image.

2.3 Combined Security Approach

Using encryption before steganographic embedding provides two security layers: the payload is protected cryptographically and its existence is concealed inside a carrier image.

2.4 Research Gap

A useful direction for future development is to combine covert communication with modern post-quantum cryptographic mechanisms, integrity verification and a modular implementation suitable for practical deployment.

CHAPTER 3 – REQUIREMENT SPECIFICATION

3.1 Functional Requirements

- Accept text or image data as input.
- Encrypt the selected payload.
- Select a suitable carrier image.
- Embed the encrypted payload into the carrier image.
- Extract the hidden payload from a stego-image.
- Decrypt and recover the original information.
- Validate the extracted information and detect invalid or corrupted data.

3.2 Non-Functional Requirements

- Security: protect confidentiality and integrity.
- Reliability: provide consistent processing and recovery.
- Usability: maintain a simple and understandable interface in the future implementation.
- Maintainability: use a modular architecture.
- Scalability: allow future cryptographic and steganographic enhancements.

3.3 Software Requirements

Python-based implementation is planned, with suitable cryptographic, image-processing and web-interface libraries to be selected during implementation. Exact libraries and versions will be finalized after development begins.

3.4 Hardware Requirements

No Raspberry Pi or other dedicated hardware is included in the current project phase. Hardware integration may be considered as a future enhancement if required.

CHAPTER 4 – SYSTEM DESIGN

4.1 Proposed Security Workflow

Input Data → Encryption → Encrypted Payload → LSB Embedding → Stego Image → Secure Transfer/Storage → Extraction → Decryption → Original Data

4.2 Data Protection

The design uses encryption as the first protection layer and steganography as the second layer. Metadata and integrity validation can be incorporated to support correct extraction and tamper detection.

4.3 Design Considerations

The design emphasizes confidentiality, covert communication, modularity and future migration toward post-quantum cryptography. The architecture is intentionally independent of a specific hardware platform at this stage.

CHAPTER 5 – SOFTWARE COMPONENTS DESCRIPTION

5.1 Cryptographic Module

Responsible for converting the input data into encrypted form and later decrypting the extracted payload. The current design can use SM4 as a symmetric encryption component, while post-quantum algorithms can be introduced in future versions.

5.2 Steganography Module

Responsible for embedding encrypted data into a carrier image using the LSB technique and extracting it during the reverse process.

5.3 Integrity and Metadata Module

Can store payload-related information such as type, size, initialization information and integrity data so that extraction can be validated.

5.4 User Interface Module

A future software interface can provide operations such as upload, encryption, embedding, extraction and download.

5.5 Logging Module

A future implementation may maintain operation logs for monitoring, debugging and accountability.

CHAPTER 6 – SYSTEM ARCHITECTURE

6.1 Architecture Overview

The proposed architecture consists of an input layer, encryption layer, steganography layer, storage/transfer layer and recovery layer. The design separates these functions so that each module can be developed and tested independently.

6.2 System Flow

User Input → Encryption Module → Encrypted Payload → LSB Steganography Module → Stego Image → Storage/Transfer → Extraction Module → Decryption Module → Recovered Data

6.3 Current Phase Status

System architecture and design have been completed for the present phase. The architecture should be treated as a proposed design rather than evidence of a completed hardware or software implementation.

CHAPTER 7 – IMPLEMENTATION

7.1 Current Implementation Status

The project is currently at the design and architecture stage. The work completed in this phase includes project definition, requirements, proposed methodology, system architecture and workflow.

7.2 Planned Implementation

The next phase will implement the encryption and steganography modules, integrate input/output handling, add validation mechanisms and develop the user interface if required.

7.3 Hardware Integration

Dedicated hardware such as Raspberry Pi is not part of the current implementation. Any hardware-based deployment will be considered only as a future enhancement and will not be presented as completed work.

CHAPTER 8 – TESTING & VALIDATION PLAN

8.1 Planned Tests

- Encryption/decryption correctness.
- Successful embedding and extraction.
- Recovery of the original payload.
- Invalid-key handling.
- Corrupted-payload detection.
- Carrier-image compatibility.
- Payload capacity and image-quality evaluation.

8.2 Validation

Quantitative measurements such as extraction accuracy and image-quality metrics may be recorded after implementation. No experimental results are claimed in this report because implementation and testing are not yet completed.

CHAPTER 9 – EXPECTED RESULTS & DISCUSSION

9.1 Expected Results

The expected outcome is a functional prototype that can encrypt sensitive information, conceal the encrypted payload inside a carrier image, recover the payload correctly and validate the extracted information.

9.2 Expected Benefits

The proposed system is expected to provide layered protection by combining data confidentiality with covert communication. Its modular design also allows future security upgrades.

CHAPTER 10 – CONCLUSION & FUTURE ENHANCEMENTS

10.1 Conclusion

The Quantum-Resistant Cyber Security Apparatus proposes a layered security approach that combines encryption and LSB image steganography. In the current academic phase, the project has progressed through analysis, requirements, system design and architecture. Full implementation, testing and practical deployment remain future work.

10.2 Future Enhancements

- Integrate standardized post-quantum cryptographic algorithms such as Kyber/ML-KEM and Dilithium/ML-DSA where appropriate.
- Improve steganographic capacity and robustness.
- Add stronger authentication and integrity protection.
- Develop a complete web or desktop interface.
- Add secure network communication and detailed logging.
- Evaluate performance, payload capacity and image quality experimentally.
- Consider portable hardware integration in a later phase.

CHAPTER 11 – REFERENCES

11.1 Reference Direction

References for the final submission should include the official or scholarly sources actually consulted for cryptography, image steganography, secure communication and post-quantum cryptography. The reference list should be finalized after verifying the sources used in the project.